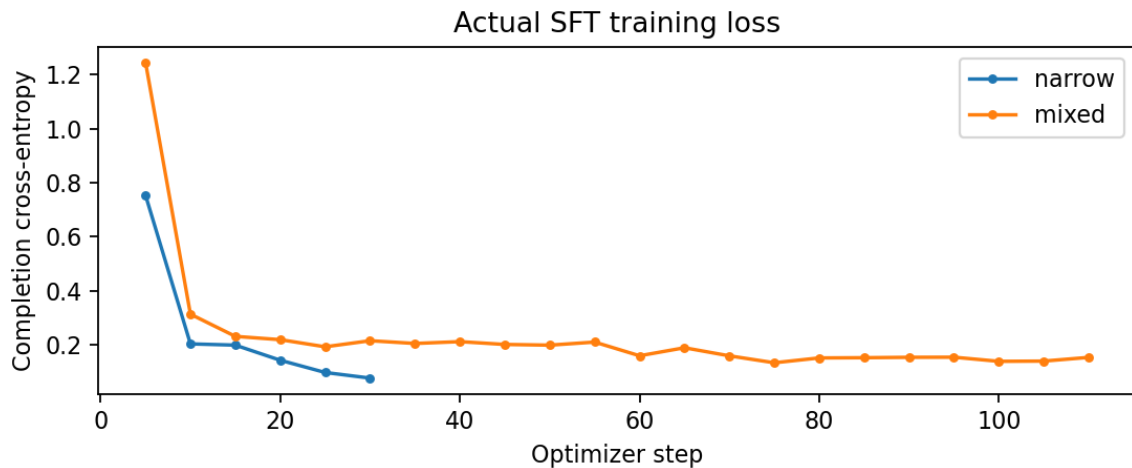


# HW5 SFT experiment report

## 1. Final method

I fine-tuned Qwen2.5-0.5B-Instruct with LoRA on 907 four-choice MMLU questions from machine learning, computer science, college mathematics, statistics, history, geography, nutrition, and elementary mathematics, using at most 120 training questions per subject [1,2]. As in the starter, MMLU test splits supply training data; dev/validation splits are held out, and questions matching either course CSV are removed before training. The final run uses two epochs, LoRA rank 16/alpha 32/dropout 0.05 on all linear layers, AdamW at peak learning rate  $1e-4$ , weight decay 0.01, cosine decay, 10% warmup, batch 1 with 16-step accumulation, gradient clipping 1, bfloat16 and seed 189 [3]. SFTTrainer minimizes completion-only cross-entropy on boxed answer letters after deterministic choice permutation; base and tuned models use the same constrained letter decoding [4]. Mixing general and specialized examples is motivated by data-composition work [5], but this one-stage experiment does not reproduce DMT.

| Model  | CS189 public (25) | Specialized holdout | General holdout |
|--------|-------------------|---------------------|-----------------|
| base   | 6/25 (24.0%)      | 30/75 (40.0%)       | 73/142 (51.4%)  |
| narrow | 12/25 (48.0%)     | 26/75 (34.7%)       | 71/142 (50.0%)  |
| mixed  | 10/25 (40.0%)     | 28/75 (37.3%)       | 72/142 (50.7%)  |



## 2. Narrow-data attempt

The narrow attempt repeatedly trained only machine-learning questions for 30 optimizer steps at learning rate  $3e-4$ ; its specialized held-out accuracy changed by -5.3 percentage points, while general accuracy changed from 51.4% to 50.0%. Its limitations are consistent with fitting a small, narrow dataset and shifting answer preferences; this comparison changes both data coverage and training budget, so it cannot isolate their causal effects.

Public accuracy is descriptive: the 25-question public set overlaps the hidden CSV. No public or hidden questions were trained on, and hidden accuracy is unavailable. The held-out MMLU results are local split evaluations, not official benchmark scores.

### 3-4. Real changes on non-CS189 inputs

Each input uses the same instruction: choose one option and return its letter in a LaTeX box. The examples below are the two shortest observed changes of each type in general held-out subjects. These illustrate individual forgetting and gains, rather than proving a broad loss or gain of an entire skill.

#### ***Forgetting 1 | cais/mmlu:elementary\_mathematics:validation***

Solve for y.  $y - 2 + 3y = 10$

A. 2   B. 3   C. 4   D. 5

Base:  $\boxed{B}$ ; final:  $\boxed{C}$ ; correct: B.

The tuned model loses a previously correct linear-equation answer; interference from the limited answer-only training mixture is one possible cause.

#### ***Forgetting 2 | cais/mmlu:elementary\_mathematics:validation***

Solve the equation  $29 = x - 17$ .

A. 12   B. -12   C. -46   D. 46

Base:  $\boxed{D}$ ; final:  $\boxed{C}$ ; correct: D.

The tuned model chooses the negative of the correct solution, suggesting that the update changed its handling of signs on this input.

#### ***Gain 1 | cais/mmlu:elementary\_mathematics:validation***

If 50% of a number is 20, what is 75% of the number?

A. 8   B. 15   C. 30   D. 45

Base:  $\boxed{B}$ ; final:  $\boxed{C}$ ; correct: C.

Exposure to other elementary-mathematics examples may have improved percentage reasoning on this held-out question.

#### ***Gain 2 | cais/mmlu:elementary\_mathematics:validation***

The expression  $-105 + (-14) + 34$  simplifies to which of the following?

A. -57   B. -75   C. -85   D. 143

Base:  $\boxed{A}$ ; final:  $\boxed{C}$ ; correct: C.

The related arithmetic training may have improved this signed-sum answer, although the output alone cannot establish the mechanism.

#### ***Kaggle status***

The 169-row submission.csv is generated and checked. Kaggle username and score screenshot are pending authenticated submission; no leaderboard score is claimed.

#### ***Sources and acknowledgement***

[1] Qwen model card. [2] cais/mmlu. [3] Hu et al., LoRA. [4] TRL SFTTrainer. [5] Dong et al., SFT data composition. OpenAI Codex assisted with code, debugging and writing; results come from actual model runs.